Week 3, Linear models: Polynomials

CAPP 30254, Machine Learning for Public Policy

Reminder: Linear decision boundaries

Recall that a linear binary classifier makes predictions:

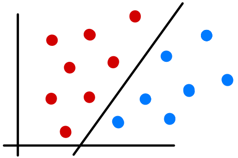
$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w}^T\mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

$\underline{w}^T \underline{x} = w_1x_1 + \dots + w_px_p + \underbrace{w_{p+1}}_{\text{bias}}$

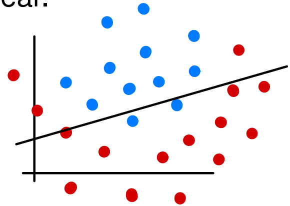
given a weight vector **w**. ~ includes bias

if not: $\underline{w}^T \underline{x} + b$

When we apply weights w_i directly to features x_i, the decision boundary will be linear.



linear



nonlinear

2D: $mx+b$

$w_1x_1+w_2$ ~ line

3D: $w_1x_1+w_2x_2+w_3$ ~ plane

4D+: hyperplane

Polynomial decision boundaries

sklearn: PolynomialFeatures

Unsurprisingly, the true boundary between two classes is often not linear.

We can create a *polynomial decision boundary* using the basic linear classification technique we've already seen, except we apply polynomial basis functions to our features (creating *new* features) before learning the weights **w**.

before: $w_i x_i$

now: $w_i \phi_i(\underline{x})$

Example: Then the prediction becomes:

x_i : feature

$\phi_i(\underline{x})$: new feature

Basis functions with degree ≤ 2

$\underline{x} \in \mathbb{R}^2$ $\quad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\phi_1 = x_1$ — linear

$\phi_2 = x_2$

$\phi_3 = x_1^2$ — quadratic

$\phi_4 = x_2^2$

$\phi_5 = x_1 x_2$ — interaction

$\phi_6 = 1$ — constant / bias

$$\hat{y} = \begin{cases} +1 & \text{if } \sum_{i=1}^{q} w_i \phi_i(\mathbf{x}) \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

$q = 6$

$w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + w_6 = \underline{w}^T \underline{\phi}(\underline{x})$

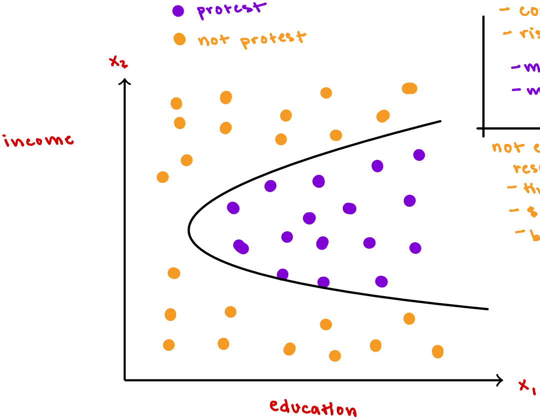
still need to learn $\underline{w}$. still a linear model!

Least squares, etc.

Example

Predict the likelihood of protesting based on income and education

Given a person's income level and education, are they likely to protest?



Income:

- connections
- risk averse
- motivation
- means

not enough resources

- time
- $$
- belief

Education:

- disengaged
- lack of access
- correlate w/ protest participation
- skepticism
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[1]

[1] Generated by GPT-4, DALL-E 3 after the prompt: “Hi, could you please make a minimalistic logo for a machine learning class...”